Simulated Agency: Consciousness as a Sparse Projection Engine

Flyxion

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Abstract

This paper presents a mathematical framework for consciousness as a *sparse projection engine*, a Bayesian heuristic system that simulates agency through constraint-driven inference. Consciousness is modeled as a function $\mathcal{C} : \mathcal{E} \to \mathcal{A}$, mapping an epistemic fiber bundle $\mathcal{E}$ to a Banach manifold of semantic trajectories $\mathcal{A}$, minimizing surprise with sparse and coherent priors. Integrating prior frameworks—scalar-vector-entropy fields, monoidal agency, quantum immanentization, and xyloarchy—the model employs stochastic differential equations (SDEs), functorial compositions, quantum projectors, and sheaf structures. Enhanced with detailed convergence proofs, information-theoretic metrics, topological analysis, and empirical predictions, the framework unifies biological, artificial, and institutional agency. Comparisons to Global Workspace Theory (GWT), Integrated Information Theory (IIT), and the Free Energy Principle highlight its novelty, while applications in neuroscience, AI, and computational psychiatry provide practical grounding. The framework prioritizes mathematical rigor and empirical testability to advance the study of consciousness.

1 Introduction

Consciousness is often conceptualized as a mysterious essence—a spark distinguishing sentient beings from machines or institutions. This paper proposes a different view: consciousness as a *sparse projection engine*, a dynamic process that sculpts actionable models of agency from complex sensory and cognitive inputs. Rather than an ontological property, consciousness emerges as a Bayesian heuristic system, mapping the *epistemic manifold* $\mathcal{E}$ to a space of *simulated agent trajectories* $\mathcal{A}$, prioritizing minimal, coherent inferences over exhaustive computation. This framework builds on prior theoretical constructs: scalar-vector-entropy fields, monoidal agency's compositional structures, quantum immanentization's retrocausal semantics, and xyloarchy's multi-scale ecological coupling, all rooted in predictive coding [1], category theory [2], ecological psychology [3], and quantum cognition [4]. It offers testable predictions for neuroscience, AI interpretability, and computational psychiatry, advancing a unified model of agency.

2 Sparse Projection Engine: A Top-Level Framework

Picture consciousness as a sculptor, selectively chiseling actionable forms from the raw stone of sensory and cognitive data. We formalize this as a function $\mathcal{C} : \mathcal{E} \to \mathcal{A}$, where:

- *Epistemic Manifold* $\mathcal{E}$: A smooth fiber bundle $\pi : \mathcal{E} \to B$, with base space B as cognitive contexts (e.g., perceptual categories) and fibers $F_b = \pi^{-1}(b)$ as belief states parameterized by coordinates $e_i \in \mathbb{R}^n$ (e.g., attention weights, memory activations). Local trivializations $\phi_b : \pi^{-1}(U) \to U \times \mathbb{R}^n$ and transition functions $g_{bb'} : U \cap U' \to \mathrm{GL}(n, \mathbb{R})$ ensure smooth belief shifts, mapping to cortical state spaces.

- *Agent Space* $\mathcal{A}$: A reflexive Banach manifold with an L_1-norm prior for sparsity, where elements $A \in \mathcal{A}$ are semantic trajectories (action policies, curves $A(t) : [0, T] \to \mathcal{A}$). A monoidal structure enables compositional agency.

The function $\mathcal{C}$ solves:

$$\mathcal{C}(P) = \arg\min_{A \in \mathcal{A}} \left[-\log p(P \mid A) + \lambda \|A\|_1 + \mu \mathcal{C}_h(A) \right],$$

where $-\log p(P \mid A)$ is the negative log-likelihood, $\lambda \|A\|_1$ enforces sparsity, and $\mu \mathcal{C}_h(A) = \mu \int_M S(A(x))\, dx$ ensures coherence via the entropy field S. Convexity guarantees a unique solution (Appendix A). This extends predictive coding [1] with sparsity, as seen in neural coding [6].

2.1 RSVP Field Triplet

The RSVP framework models consciousness as three coupled fields over a spacetime manifold $M = \mathbb{R} \times \mathbb{R}^3$ with Minkowski metric $g_{\mu\nu} = \mathrm{diag}(-1, 1, 1, 1)$:

1. *Scalar Coherence Field* Φ: Aligns predictions with sensory inputs, representing cortical semantic load.
2. *Vector Inference Flow* $\vec{v}$: Directs belief updates, corresponding to attentional shifts in cognitive processing.
3. *Entropy Field* S: Quantifies prediction error, reflecting the degree of surprise in the system.

These evolve via Itô SDEs:

$$d\Phi_t = \left[\nabla \cdot (D\nabla\Phi_t) - \vec{v}_t \cdot \nabla\Phi_t + \lambda S_t\right] dt + \sigma_\Phi dW_t,$$

$$d\vec{v}_t = \left[-\nabla S_t + \gamma \Phi_t \vec{v}_t\right] dt + \sigma_v dW_t',$$

$$dS_t = \left[\delta \nabla \cdot \vec{v}_t - \eta S_t^2\right] dt + \sigma_S dW_t'',$$

derived from a Lagrangian (Appendix B). Boundary conditions set $\Phi, S, \vec{v} \to 0$ at sensory limits. Well-posedness is proven via Lipschitz continuity (Appendix B). RSVP captures how consciousness balances model alignment, updates, and uncertainty, mapping to neural processes or AI attention mechanisms.

2.2 Empirical Predictions

The RSVP framework yields the following predictions:

1. *Neural*: Sparse projections in RSVP fields correlate with enhanced neural synchrony in attention tasks, observable via EEG.
2. *Behavioral*: Sparse inference reduces reaction times in decision-making tasks, testable through psychophysical experiments.
3. *Clinical*: Disruptions in entropy field S dynamics are associated with cognitive disorders, measurable via fMRI connectivity patterns.
4. *Developmental*: Developing brains exhibit less sparse inference patterns, observable in attention task performance across age groups.

2.3 Comparison to Existing Theories

RSVP generalizes the Free Energy Principle [1] with field dynamics, predicts finer neural synchrony than GWT [7], and offers lower complexity than IIT's ϕ [8].

3 Monoidal Agency

Agency is an orchestra of subagents—neurons, AI layers, or corporate roles—coordinating for unified action. In category $\mathcal{C}$, objects are inference states, and morphisms are sparse projectors. An agent is a monoid $(M, \cdot, e) \in \mathbf{Mon}(\mathcal{C})$, with M as subagent states, $\cdot$ as decision chaining, and e as homeostasis. A functor $\mathcal{F} : \mathbf{Env} \to \mathbf{Mon}(\mathcal{C})$ maps environmental states (sensorimotor structures, affordance morphisms) to monoids, with natural transformations modeling context shifts. A sheaf $\mathcal{S}_C$ patches inferences, with $H^1(\mathcal{C}, \mathcal{S}_C)$ quantifying dissonance (Appendix C). This links to modular cognitive architectures.

4 Quantum Immanentization

4.1 Background and Motivation

Consciousness often feels like a narrative constructed retroactively, weaving ambiguous sensory inputs into coherent meaning. For example, when you hear a faint sound in the dark, your mind might retroactively interpret it as a threat, shaping your actions. This process, termed *quantum immanentization*, draws inspiration from quantum cognition [4], where cognitive states exhibit non-classical properties like superposition and interference. Unlike physical quantum systems, we use a Hilbert space formalism to model semantic ambiguity and memory-driven stabilization, building on recursive tiling of semantic trajectories. This section formalizes how consciousness stabilizes agency through retrocausal projections, connecting to predictive processing [9] and neural feedback mechanisms [7].

4.2 Mathematical Formalism

We model cognitive states in a separable Hilbert space $\mathcal{H}$, with an orthonormal basis $\{|\psi_i\rangle\}$ representing discrete semantic concepts (e.g., "threat," "reward," "self"). An observer's memory, encoded in neural weights, induces a projection operator:

$$P_{\text{mem}} = \sum_i w_i |\psi_i\rangle\langle\psi_i|,$$

where $w_i \in [0, 1]$ reflects memory strength, derived from synaptic weights or AI attention scores. For an observation $O \in \mathcal{H}$ (e.g., sensory input), the *meaning* is constructed as:

$$\text{Meaning}(O) = \frac{P_{\text{mem}} O P_{\text{mem}}}{\|P_{\text{mem}} O P_{\text{mem}}\|},$$

projecting O onto memory-consistent subspaces. The state evolves via a retrocausal integral:

$$|\psi(t)\rangle = \int_{-\infty}^{t} U(t, \tau) P_{\text{mem}} |\psi(\tau)\rangle d\tau,$$

where $U(t, \tau)$ is a unitary propagator derived from neural dynamics (e.g., thalamocortical feedback loops). Decoherence, a challenge in physical quantum systems, is mitigated by recurrent neural feedback, with decoherence times estimated at 10–100 ms based on thalamocortical oscillation frequencies [4]. This ensures semantic coherence in noisy biological systems, unlike quantum computers requiring cryogenic conditions.

4.3 Empirical Connections

The formalism predicts *order effects* in decision-making, where the sequence of choices alters outcomes due to interference in $\mathcal{H}$ [4]. For example, in a sequential choice task (e.g., judging ambiguous images), the order of presentation should modulate reaction times, testable via psychophysical experiments. In AI, P_{mem} maps to attention mechanisms in Transformers, where token weights act as projectors. This connects to recursive semantic tiling, where fields are stabilized.

4.4 Neurobiological and AI Applications

The quantum immanentization framework yields the following applications:

1. Increased gamma synchrony (30–100 Hz) during memory-driven tasks, measurable via EEG.

2. Lower perplexity in LLMs with sparse P_{mem}, testable via ablation studies.

This section unifies retrocausal semantics with the sparse projection engine, explaining how consciousness constructs agency from ambiguous inputs.

5 Xyloarchy

5.1 Background and Motivation

Consciousness does not exist in isolation but is embedded in a multi-scale ecological web, from neurons to social institutions. The *xyloarchy* framework, inspired by ecological psychology [3], models this as a hierarchical system of affordances and constraints across temporal and spatial scales. For example, a single neuron's firing influences a decision, which shapes behavior, which impacts social dynamics. Xyloarchy formalizes this cross-scale coupling, integrating RSVP fields to capture how consciousness emerges from ecological interactions.

5.2 Mathematical Formalism

We model the ecological hierarchy as a topological space X, representing nested layers (neural, behavioral, social). A sheaf $\mathcal{S} : \mathbf{Open}(X) \to \mathbf{Vec}$ assigns to each open set $U \in \mathbf{Open}(X)$ a vector space $\mathcal{S}(U)$ of affordances (action possibilities) and constraints (environmental limits). Restriction maps $\rho_{UV} : \mathcal{S}(U) \to \mathcal{S}(V)$ for $V \subseteq U$ model cross-scale interactions, ensuring compatibility via transition functions ϕ_{UV}. A time-indexed sheaf $\mathcal{S}_t$ captures temporal scales:

1. *Fast scale* (100 ms): Attention and sensory processing.
2. *Medium scale* (seconds): Narrative coherence and decision-making.
3. *Slow scale* (years): Belief and identity formation.

The first cohomology $H^1(X, \mathcal{S})$ quantifies cognitive conflicts (e.g., misaligned affordances), relevant to disorders like ADHD or schizophrenia. For example, high H^1 values indicate disrupted cross-scale coherence, observable in neural dysconnectivity.

5.3 Empirical and Clinical Applications

The xyloarchy framework yields the following predictions:

1. Xyloarchy predicts sparse neural activation in attention tasks, with fewer active nodes in efficient cognitive systems, testable via fMRI [10].
2. In clinical settings, disrupted sheaf morphisms (e.g., weak ρ_{UV}) correlate with cognitive disorders, measurable via EEG coherence metrics, with schizophrenia manifesting as high H^1 due to fragmented affordance integration, suggesting novel diagnostic criteria.

5.4 Connection to Prior Work

This framework extends ecological layering, embedding RSVP fields in a multi-scale topology. It supports the sparse projection engine by grounding agency in ecological feedback, from neural circuits to social systems.

6 Information-Theoretic Analysis

6.1 Background and Motivation

Consciousness can be viewed as an information bottleneck, compressing high-dimensional sensory inputs into low-dimensional actionable representations. This aligns with sparse coding in neural systems [6] and attention mechanisms in AI [11]. The *information-theoretic analysis* quantifies how the sparse projection engine optimizes information flow between $\mathcal{E}$ and $\mathcal{A}$, drawing on predictive processing [1] and rate-distortion theory [12].

6.2 Mathematical Formalism

We define mutual information between the epistemic manifold $\mathcal{E}$ and agent space $\mathcal{A}$:

$$I(\mathcal{E}; \mathcal{A}) = \iint p(e,a) \log\left(\frac{p(e,a)}{p(e)p(a)}\right) de\, da,$$

where $p(e,a)$ is the joint probability density, $p(e)$ and $p(a)$ are the marginal density functions, and the integral is over the domains of $e \in \mathcal{E}$ and $a \in \mathcal{A}$. This measures information shared between sensory inputs and agent trajectories. To optimize sparsity, we apply rate-distortion theory, minimizing distortion:

$$D = \mathbb{E}[\|P - \mathcal{C}(P)\|^2],$$

subject to a rate constraint R (information transmitted). The sparse projection $\mathcal{C}$ solves:

$$\min_{\mathcal{C}} D + \beta I(\mathcal{E}; \mathcal{A}),$$

where β balances distortion and rate, aligning with the L_1-norm in $\mathcal{C}(P)$. This formalism quantifies consciousness as an efficient encoder, compressing epistemic states into sparse agent models.

6.3 Empirical Predictions

The information-theoretic formalism yields the following predictions:

1. High mutual information $I(\mathcal{E}; \mathcal{A})$ predicts efficient cognitive processing, testable via EEG entropy in attention tasks.
2. Low distortion D in AI systems correlates with interpretable LLM outputs, measurable via perplexity reduction.
3. Clinically, reduced mutual information may indicate disorders like autism, where sensory integration is impaired, testable via fMRI connectivity.

6.4 Connection to Prior Work

This information-theoretic analysis extends the entropy field S, which quantifies prediction error, by embedding it within a broader framework of information optimization across cognitive scales. The entropy field S builds on the Free Energy Principle [1], which models cognition as minimizing variational free energy to reduce prediction error. Unlike the Free Energy Principle's focus on global energy minimization, our framework incorporates field dynamics through the coupled evolution of S, Φ, and $\vec{v}$, enabling spatially and temporally localized surprise minimization, as seen in the SDEs of Section 2.1. This dynamic approach aligns with hierarchical predictive coding, where multi-scale predictions refine sensory processing [9].

The emphasis on sparsity in $\mathcal{C}(P)$ draws directly from sparse coding principles in neural systems [6], further formalized by Elad's work on sparse and redundant representations [17]. Elad's framework demonstrates how overcomplete dictionaries enable efficient signal representation with minimal active components, a concept mirrored in our L_1-norm regularization for semantic trajectories in $\mathcal{A}$. This sparsity reduces computational complexity in biological and artificial systems, as evidenced by efficient neural coding in V1 and interpretable AI outputs. For example, sparse representations in convolutional neural networks align with our model's compression of epistemic states, enhancing robustness to noise and improving generalization.

In AI, the mutual information $I(\mathcal{E}; \mathcal{A})$ connects to attention mechanisms in Transformer architectures [11], where attention weights act as sparse selectors of relevant input features, analogous to the vector inference flow $\vec{v}$. This parallelism suggests that our framework's optimization of $I(\mathcal{E}; \mathcal{A})$ can inform the design of interpretable AI systems, where sparsity constraints reduce overfitting and enhance semantic coherence. Additionally, the rate-distortion framework [12] provides a theoretical backbone for balancing information fidelity (low D) with compression (low R), a trade-off central to our model's efficiency in mapping $\mathcal{E}$ to $\mathcal{A}$. This connects to ecological models of cognition [3], where affordance-driven compression filters sensory inputs to prioritize actionable outcomes.

The modular compression aspect links to hierarchical cognitive architectures, where submodules process specific sensory or cognitive domains, akin to the sheaf structures in xyloarchy (Section 5). This modularity ensures that information bottlenecks are distributed across scales, from neural circuits to social systems, aligning with ecological psychology's emphasis on context-dependent perception. By integrating these frameworks, our analysis provides a quantitative foundation for evaluating sparse projections, offering a unified perspective on consciousness as an efficient, multi-scale information processor.

7 Dynamical Systems and Topological Analysis

7.1 Background and Motivation

Consciousness exhibits dynamic stability, with stable states (e.g., focused attention) and transitions (e.g., insight moments). The *dynamical systems and topological analysis* frames conscious states as attractors in RSVP phase space, with transitions modeled as bifurcations. Topological tools, like persistent homology, reveal stable features of cognitive processes, drawing on dynamical systems theory [13] and computational topology [14]. This connects to multi-scale ecological dynamics.

7.2 Mathematical Formalism

We model RSVP fields as a dynamical system on the phase space $(\Phi, \vec{v}, S)$. Stable conscious states are fixed-point attractors, where:

$$\frac{d\Phi}{dt} = 0, \quad \frac{d\vec{v}}{dt} = 0, \quad \frac{dS}{dt} = 0.$$

Bifurcations occur when parameters (e.g., λ, γ) cross critical thresholds, triggering transitions (e.g., attention shifts). For example, a saddle-node bifurcation may model a shift from inattention to focus. The Jacobian of the SDE system determines stability:

$$J = \begin{bmatrix} D\nabla^2 - \vec{v} \cdot \nabla & -\nabla\Phi & \lambda \\ \gamma\vec{v} & \gamma\Phi & -\nabla \\ -\eta S & \delta\nabla & -2\eta S \end{bmatrix}.$$

Eigenvalues of J indicate attractor stability. Topologically, we apply persistent homology to RSVP fields, computing Betti numbers β_k to quantify "holes" in cognitive space (e.g., β_1 for disconnected affordances). Persistent features correspond to stable conscious states.

7.3 Empirical Predictions

The dynamical systems framework yields the following predictions:

1. Attractor stability predicts sustained attention, testable via EEG coherence in sustained attention tasks.
2. Bifurcations correlate with insight moments, measurable via reaction time spikes.
3. High β_1 values indicate cognitive fragmentation, relevant to schizophrenia, testable via fMRI connectivity.

7.4 Connection to Prior Work

This analysis extends RSVP field dynamics and ecological scale interactions, providing a rigorous framework for studying consciousness as a dynamic, topological phenomenon.

8 Sparse Projection Operator

The operator $\mathcal{A}(x) = \arg\min_{T\in\mathcal{T}} \left[\int_T S(\tau)\, d\tau + \beta\|T\|_0\right]$ unifies the framework, with Runge-Kutta simulation (Appendix F).

9 Sparse Token Inference as a Model of Simulated Agency

9.1 Background and Motivation

The sparse projection engine posits that consciousness emerges from minimal, coherent inferences rather than exhaustive generative models. Recent advances in token-based inference, as demonstrated by Beyer et al. [15], provide a concrete realization of this principle. Their 1D tokenizer framework generates complex outputs (e.g., images) via sparse, feedback-guided operations on a compressed latent space, without requiring large-scale retraining. This aligns with the Simulated Agency framework's claim that agency arises from constrained inference in latent dynamics, not from massive parameterized models. By mapping token operations to RSVP's field dynamics, we formalize how sparse token inference embodies the sparse projection engine, connecting to predictive processing [1] and recursive semantic trajectories.

9.2 Latent Representation as Sparse Projection

Consider a perceptual input I (e.g., an image) encoded by a pretrained tokenizer into a sequence of $K = 32$ discrete latent tokens:

$$\mathbf{z} = \mathrm{Tok}(I) = [z^{(1)}, z^{(2)}, \ldots, z^{(K)}], \quad z^{(k)} \in \mathcal{D}, \quad |\mathcal{D}| = 4096,$$

where $\mathcal{D}$ is a discrete codebook. This sequence represents a sparse projection of the epistemic manifold $\mathcal{E}$ into a low-dimensional latent space $\mathcal{D}^K$, analogous to the scalar coherence field $\Phi : \mathcal{M} \to \mathcal{D}^K$. Each token $z^{(k)}$ encapsulates nonlocal semantic content (e.g., lighting, texture), enabling sparse updates to influence global properties, as observed in Beyer et al. [15]. This compression mirrors the L_1-norm sparsity in $\mathcal{C}(P)$, reducing computational complexity while preserving semantic coherence.

9.3 Discrete Edit Operators as Local Field Perturbations

Sparse inference operates via localized edit operators on the token sequence. Define a single-token copy-paste operator for a target latent $\mathbf{z}_{\text{target}}$ and reference latent $\mathbf{z}_{\text{ref}}$:

$$\mathcal{T}_k^{\text{swap}}(\mathbf{z}_{\text{target}}, \mathbf{z}_{\text{ref}}) = \left[z_{\text{target}}^{(1)}, \ldots, z_{\text{target}}^{(k-1)}, z_{\text{ref}}^{(k)}, z_{\text{target}}^{(k+1)}, \ldots, z_{\text{target}}^{(K)} \right].$$

This induces a local perturbation in the scalar field:

$$\delta\Phi_k = \Phi_k^{\text{ref}} - \Phi_k^{\text{target}},$$

where Φ_k represents the semantic contribution of the k-th token. Such perturbations, as shown in Beyer et al. [15], modify global image properties (e.g., color tone, sharpness), aligning with RSVP's vector inference flow $\vec{v}$, which directs updates to minimize the entropy field S. This mechanism supports the Simulated Agency hypothesis that agency emerges from localized, coherent updates in a sparse latent space.

9.4 Feedback-Guided Optimization as Projective Inference

The tokenizer framework optimizes latent tokens using a feedback coherence functional, such as CLIP-based similarity [16]. For a decoded image $\mathrm{Dec}(\mathbf{z})$ and a target text prompt T, the optimization is:

$$\max_{\mathbf{z}} \mathcal{L}(\mathbf{z}) = \ell(\mathrm{Dec}(\mathbf{z}), T),$$

where ℓ is a similarity metric (e.g., cosine similarity in CLIP's embedding space). Gradient-based updates proceed as:

$$\mathbf{z}^{(t+1)} = \mathbf{z}^{(t)} + \eta \cdot \nabla_{\mathbf{z}} \mathcal{L}(\mathbf{z}^{(t)}),$$

with vector quantization handled via a straight-through estimator [15]. In RSVP terms, this corresponds to maximizing a coherence functional:

$$\max_{\Phi} \mathcal{C}[\Phi] = \int_{\mathcal{M}} w(x) \cdot \mathcal{K}(\Phi(x), T(x))\, dx,$$

where $\mathcal{K}$ is a local coherence kernel (e.g., cosine similarity) and $w(x)$ weights semantic relevance, aligning with the entropy field S. This process mirrors the projective inference in $\mathcal{C} : \mathcal{E} \to \mathcal{A}$, where feedback minimizes surprise while maintaining sparsity.

9.5 Empirical Predictions and Applications

The tokenizer framework predicts that sparse token edits yield significant semantic changes, testable in AI systems by measuring output fidelity (e.g., Fréchet Inception Distance) after single-token swaps. In neuroscience, analogous sparse updates in neural representations (e.g., prefrontal cortex) should correlate with gamma synchrony (30–100 Hz) during attention tasks, measurable via EEG. Clinically, disruptions in sparse token-like inference may manifest as cognitive inflexibility in disorders like autism, testable via fMRI connectivity. In AI, the framework suggests low-cost generative models for scenario completion, where token sequences encode personas or contexts, aligning with recursive semantic trajectories.

9.6 Future Directions

The sparse token inference framework opens the following research avenues:

1. *Social Agents*: Embedding personas in token sequences to simulate multi-agent interactions, testable via dialogue coherence metrics.
2. *Scenario Completion*: Generating linguistic or visual narratives via sparse token edits, without retraining, measurable by perplexity reduction.
3. *Alignment Heuristics*: Optimizing value coherence in AI via CLIP-guided token feedback, reducing computational costs.
4. *Low-Cost Reasoning*: Developing agents that leverage RSVP field projections and CLIP guidance for conceptual reasoning, testable in tasks like analogy solving.

10 Conclusion

The *sparse projection engine* framework redefines consciousness as a dynamic process that simulates agency through minimal, coherent inferences. By integrating RSVP's field dynamics, monoidal agency's compositional structures, quantum immanentization's retrocausal semantics, xyloarchy's ecological layering, and sparse token inference, it provides a unified model of agency across biological, artificial, and institutional systems. The framework's mathematical rigor—formalized through fiber bundles, SDEs, category theory, information-theoretic metrics, and token-based projections—offers a robust foundation for studying consciousness, while its empirical predictions (e.g., gamma synchrony, sparse attention, clinical diagnostics) bridge theory to experiment. Compared to GWT, IIT, and the Free Energy Principle, it emphasizes sparsity and multi-scale dynamics, predicting finer-grained neural and behavioral outcomes. Applications in AI interpretability, computational psychiatry, and collective intelligence demonstrate its practical impact.

Future research should prioritize EEG and fMRI validation of RSVP predictions, computational simulations in SymPy or Julia, and AI implementations to test sparse projection in LLMs and token-based systems. Clinical studies could explore therapeutic interventions (e.g., tDCS for RSVP modulation), while philosophical inquiries might address the ethical implications of simulated agency in AI and institutions. By synthesizing mathematical sophistication with empirical grounding, this framework lays the groundwork for a transformative research program in consciousness studies.

A Convergence Proofs for Sparse Projection Operator

To ensure the sparse projection operator $\mathcal{C}(P) = \arg\min_{A\in\mathcal{A}} [-\log p(P \mid A) + \lambda\|A\|_1 + \mu\mathcal{C}_h(A)]$ is well-defined, we prove existence and uniqueness. The objective $\mathbb{S}(A \mid P) = -\log p(P \mid A) + \lambda\|A\|_1 + \mu \int_M S(A(x))\, dx$ is defined on the reflexive Banach manifold $\mathcal{A}$. Assume $p(P \mid A)$ is log-concave (e.g., Gaussian likelihood), ensuring $-\log p(P \mid A)$ is convex. The L_1-norm is convex, and $\mathcal{C}_h(A)$ is convex since S is non-negative and continuous. Strict convexity follows from differentiability of $-\log p(P \mid A)$. By the Weierstrass Theorem, a unique minimizer exists in compact $\mathcal{A}$. For numerical optimization, gradient descent with step size $\eta < 1/L$ (where L is the Lipschitz constant) converges to the global minimum. This mirrors sparse coding in neural networks [6], where sparsity minimizes computational load. The coherence term $\mathcal{C}_h(A)$ links to the entropy field, ensuring alignment with predictive processing [1].

B RSVP Field Equations and Variational Derivation

The RSVP fields evolve over $M = \mathbb{R} \times \mathbb{R}^3$ with metric $g_{\mu\nu}$. The Lagrangian is:

$$\mathcal{L} = \frac{1}{2} g^{\mu\nu}\partial_\mu\Phi\partial_\nu\Phi + \alpha\vec{v} \cdot \nabla\Phi + \beta S\nabla \cdot \vec{v} - \frac{\eta}{2}S^2 + \lambda\Phi S.$$

Euler-Lagrange equations yield the SDEs. Well-posedness requires Lipschitz drift and bounded noise, satisfied under $\sigma_\Phi, \sigma_v, \sigma_S < \infty$. Semantic flux conservation follows from $\nabla \cdot (\Phi\vec{v}) \approx 0$ in steady states. Boundary conditions reflect sensory limits, where cognitive processing saturates. This extends RSVP to stochastic dynamics, aligning with predictive coding [9].

C Coherence Diagrams for Monoidal Agency

The functor $\mathcal{F} : \mathbf{Env} \to \mathbf{Mon}(\mathcal{C})$ maps **Env** (objects: sensorimotor states; morphisms: affordance transitions) to monoids. For morphisms $f, g \in \mathbf{Env}$, $\mathcal{F}(f \circ g) = \mathcal{F}(f) \cdot \mathcal{F}(g)$, and $\mathcal{F}(\mathrm{id}_E) = e_E$. Natural transformations $\eta : \mathcal{F} \to \mathcal{F}'$ model context shifts, satisfying commutative diagrams. The sheaf $\mathcal{S}_C$ patches inferences, with $H^1(\mathcal{C}, \mathcal{S}_C)$ quantifying conflicts, akin to modular cognitive architectures.

D Stochastic Dynamics of RSVP Fields

The Itô SDEs include uncorrelated Brownian sheets, with $\sigma_\Phi, \sigma_v, \sigma_S$ tuned to neural noise levels (0.01–0.1). Numerical stability is ensured via 4th-order Runge-Kutta integration, with step size $dt < 1/\max(\lambda, \gamma, \delta, \eta)$.

E Quantum Immanentization and Hilbert Structures

The Hilbert space $\mathcal{H}$ has basis $|\psi_i\rangle$ as semantic concepts, with P_{mem} constructed from neural weights. Decoherence times (10–100 ms) are estimated from thalamocortical oscillations, ensuring coherence in noisy systems [4]. The retrocausal integral links to recursive semantic tiling, predicting order effects in choice tasks.

F RSVP Simulation and Validation

```
import numpy as np
def simulate_rsvp(T, grid_shape, params):
    phi = np.zeros(grid_shape)
    v = np.zeros(grid_shape + (2,))
    S = np.ones(grid_shape)
    dt = params.dt
    for t in range(T):
        grad_phi = np.gradient(phi)
        div_v = np.divergence(v)
        k1_phi = params.D * laplacian(phi) - np.dot(v, grad_phi) + params.
            lamb * S
        k1_v = -np.gradient(S) + params.gamma * phi[..., np.newaxis] * v
        k1_S = params.delta * div_v - params.eta * S**2
        phi += k1_phi * dt + params.sigma_phi * np.random.normal(0, np.sqrt
            (dt), grid_shape)
        v += k1_v * dt + params.sigma_v * np.random.normal(0, np.sqrt(dt),
            grid_shape + (2,))
        S += k1_S * dt + params.sigma_S * np.random.normal(0, np.sqrt(dt),
            grid_shape)
    return phi, v, S

def validate_rsvp_simulation():
    params = {'D': 0.1, 'lamb': 0.5, 'gamma': 0.2, 'delta': 0.3, 'eta':
        0.1,
              'sigma_phi': 0.01, 'sigma_v': 0.01, 'sigma_S': 0.01, 'dt':
                 0.01}
    phi, v, S = simulate_rsvp(1000, (100, 100), params)
    mass_initial = np.sum(S)
    phi, v, S = simulate_rsvp(1000, (100, 100), params)
    mass_final = np.sum(S)
    assert np.allclose(mass_initial, mass_final, atol=1e-5)
```

G Comparison to Recursive Inference Architectures

Recent recursive inference architectures, such as those proposed by Cheng et al. (2025) [18], model cognitive processes through iterative reasoning, memory anchoring, and belief synthesis, building on established algorithms like chain-of-thought and active inference. The *sparse projection engine* framework similarly produces recursive behaviors but derives them from the dynamics of RSVP fields, formalized using category theory and sheaf structures. Specifically, these dynamics exhibit emergent structures resembling our prior RAT architecture, TARTAN framework, and Yarncrawler tiling:

1. *Yarncrawler-like Tiling*: The vector inference flow $\vec{v}$ guides recursive partitioning of the epistemic manifold $\mathcal{E}$, creating semantic tiles with Φ-coherence and S-weighted uncertainty, akin to trajectory-aware tiling in recursive inference.
2. *RAT-like Recurrence*: Stable attractors in the Φ-S manifold, where $\frac{d\Phi}{dt} \to 0$ and ∇S is stable, anchor trajectories for sparse recurrence, mirroring iterative refinement in our RAT architecture.
3. *TARTAN-like Clustering*: Semantic perturbations in RSVP fields, driven by $\vec{v}$ and constrained by S, form belief graphs with nodes as coherent states and edges as inference transitions, similar to TARTAN's annotated clustering.

These emergent behaviors align with the recursive traversal, confidence thresholding, and belief graph synthesis described in recent work [18]. While CLIO employs iterative reasoning and descriptive terminology to model cognitive dynamics, our framework uses the mathematical structure of RSVP fields to generate recursive inference, providing a distinct perspective on agency across neural and AI systems.

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